# Parktown Boys' High School

## Geography Department

## Grade 12 June Exam

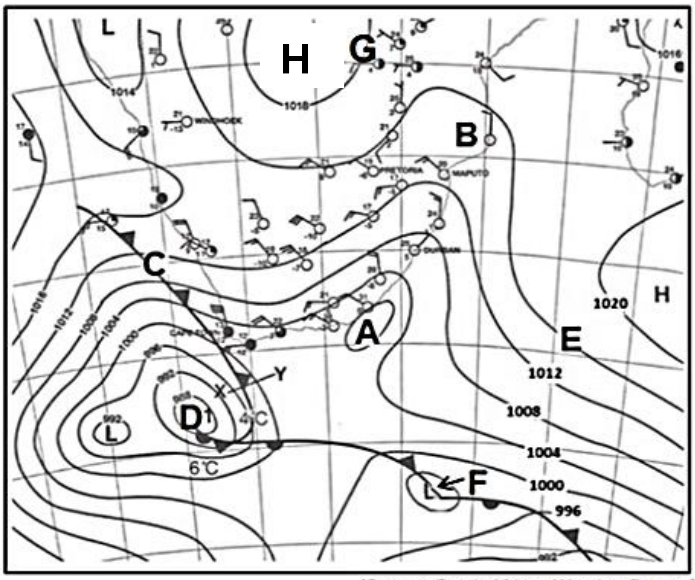
## MEMO

# SECTION A: THEORY

**QUESTION 1** **CLIMATOLOGY**

**Question 1.1** **Synoptic charts**

Choose the correct word(s) from those given in brackets. Write only the word(s) next to the question numbers (1.1.1 to 1.1.5) in the ANSWER BOOK.




[Source: South African Weather Bureau]

1.1.1 Pressure cell **A** is a (Cut off / Coastal) low pressure.

1.1.2 Area **B** is experiencing (Northerly/ Southerly) winds.

1.1.3 The system at **F** is a (Tropical Cyclone / Mid latitude cyclone).

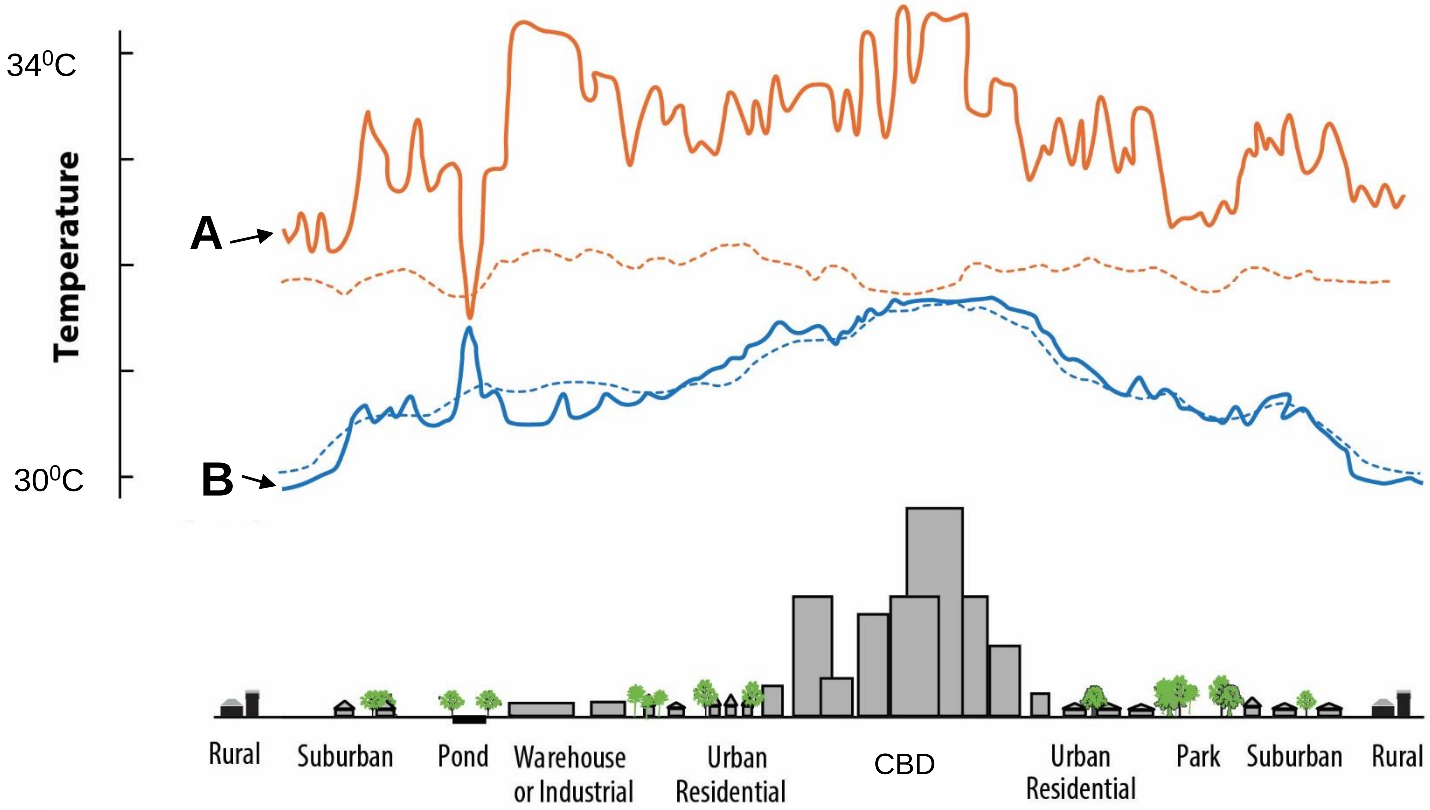
1.1.4 The pressure cell at **H** is a (Thermal Low pressure / Kalahari High pressure).

1.1.5 The synoptic chart is representing a (Summer / Winter) season.

(5x1) (5)

**Question 1.2** **City climates**

Choose which graph best matches the description given. Write down the question number (1.2.8) and A or B only (1.2.8 A)




1.2.1 Graph …shows the time when artificial heat is mostly given off.

1.2.2 There is an increase in this graph due to water holding heat longer.

1.2.3 Roof top gardens and additional vegetation will cause a huge difference in which graph?

1.2.4 A Kalahari high-pressure cell has the most effect on which graph?

1.2.5 Which graph represents an urban heat island?

(5x1) (5)

**[10]**

**Question 1.1**

1.1.1 Coastal (1)

1.1.2 Northerly (1)

1.1.3 Mid latitude Cyclone (1)

1.1.4 Kalahari High pressure (1)

1.1.5 Winter (1) (5x1) (5)

**Question 1.2**

1.2.1 A (1)

1.2.2 B (1)

1.2.3 A (1)

1.2.4 B (1)

1.2.5 A (1) (5x1) (5)

**Question 1.3** **Mid-Latitude Cyclones**

**ARTICLE PRESENTED**

1.3.1 Name the weather phenomenon that a cold front is part of. (1x1) (1)

Mid latitude cyclone / depression (1)

1.3.2 Give one phrase from the passage that states that snowstorms are not a common occurrence in South Africa. (1x1) (1)

" disaster area yesterday after the worst snow in decades " (1)

" It was the worst snowstorm in 70 years,"(1)

1.3.3 Where does the extreme cold air of this system originate (come from)? (1x1) (1)

the poles (1)

Polar air (1)

Polar easterly winds(1)

1.3.4 Fully explain how a cold front causes snow to develop over South Africa? (2x2) (4)

<u>Very cold air mass</u> moves over the <u>interior pushing the moist air up</u> (2)

Warm moist air from the eastern coastal area is forced to <u>rise quickly</u> (2)

Condensation does not take place as the warm water vapour freezes due to the cold air mass.(2)

Sublimation occurs as the cold air mass forces the moist air to rise (2)

Cold front moving faster than the warm front (2)

Steep pressure gradient force causes rapid upliftment of air(2)

1.3.5 In a paragraph of approximately EIGHT lines, discuss how a cold front can increase food insecurity in South Africa. (4x2) (8)

Cold front can cause black frost and kill the vegetation / soil freezes (2)

Stock has no fodder therefore no meat therefore no food (2)

Cold front can kill stock – no food (2)

Cold front will wash away crops due to flooding, no food (2)

Flooding can cause erosion of soil / nutrients lost and then production of crops limited therefore food insecurity (2)

Valuable nutrients in the soil can be frozen. (2)

People can lose homes need to move loss of income no food (2)

Food prices will increase people cannot afford food lack of food (2)

People will lose jobs on farms as yield per hector will decrease (2)

Farmers will need to spend money fixing infrastructure due to flooding or wind damage and the not having money to pay workers (2)

People who have lost jobs won’t be able to afford food / food insecurity (2)

**[15]**

**Question 1.4** **The figure shows the development of a moisture front.**


1.4.1 What is a *moisture front*? (1x2) (2)

Line separating moist air from dry air (2)

A boundary separating two different types of air masses (2)

1.4.2 In which season does a moisture front mainly develop? (1x1) (1)

summer season (1)

1.4.3 Name the pressure cells that cause the winds at **A** and **B** respectively. (2x1) (2)

**MUST HAVE "South" otherwise ZERO**

**A** – South Atlantic High pressure / anticyclone(1)

**B** – South Indiana High pressure / anticyclone (1)

1.4.4 How does the air from pressure cells **A** and **B** influence the development of the moisture front? (2x2) (4)

Different air masses don't mix, they form a front. (2)

The Warm moist air from the South Indian HP rises rapidly (2)

The cold air SW from the south Atlantic HP wedges underneath the warm air NE ahead of it (2)

Air travels from a HP to a LP (2)

There is a trough of LP over the interior (2)

Cold dry SW winds blow towards the LP over the interior (2)

Warm moist NE blow towards the LP over the interior (2)

The cold dry SW air forces the warm moist NE air to rise (2)

Cumulonimbus clouds develop on the eastern side of the moisture front (2)

1.4.5 Explain how this system can be both a curse and a blessing to farmers. (3x2) (6)

ANSWER RATIO 2:1

**Blessing:**

Brings rain / increases water in dams' reservoirs / underground water (2)

Increase water, increase yield, more crops grow (2)

increase sales / increase income (2)

less money spent on irrigation and water pumps (2)

more food security (2)

increase water of paddocks / better stock growth (2)

hail can bring minerals to soil (2)

Lightning can assist in development of nitrates.

**A curse:**

Can cause flooding of land / infrastructure (2)

Soil erosion / short- and long-term damage (2)

Loss of crops / stock (2)

Hail can cause damage of crops nothing to sell/ stock (2)

Loss of income (2)

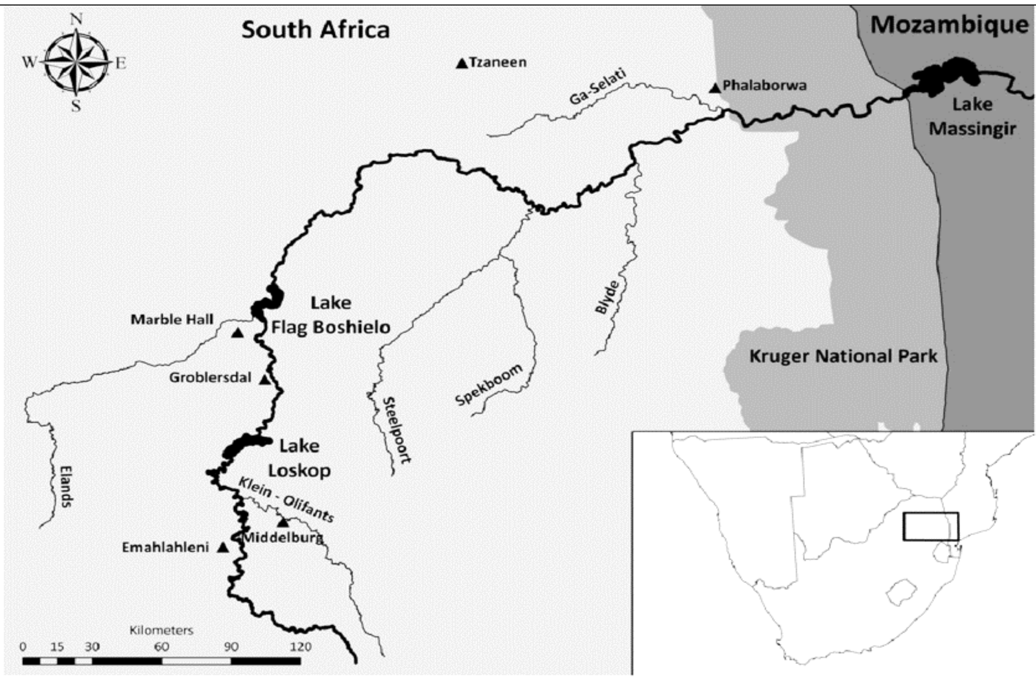
**[15]**

**QUESTION 1 TOTAL: 40 MARKS**

**QUESTION 2** GEOMORPHOLOGY

**Question 2.1.** Stream order

Refer to the map showing the Olifants river system. Complete the statements in COLUMN A with the options in COLUMN B. Write only **X** or **Y** next to the question numbers (2.1.1 to 2.1.5) in the ANSWER BOOK, e.g. 2.1.6 Y.



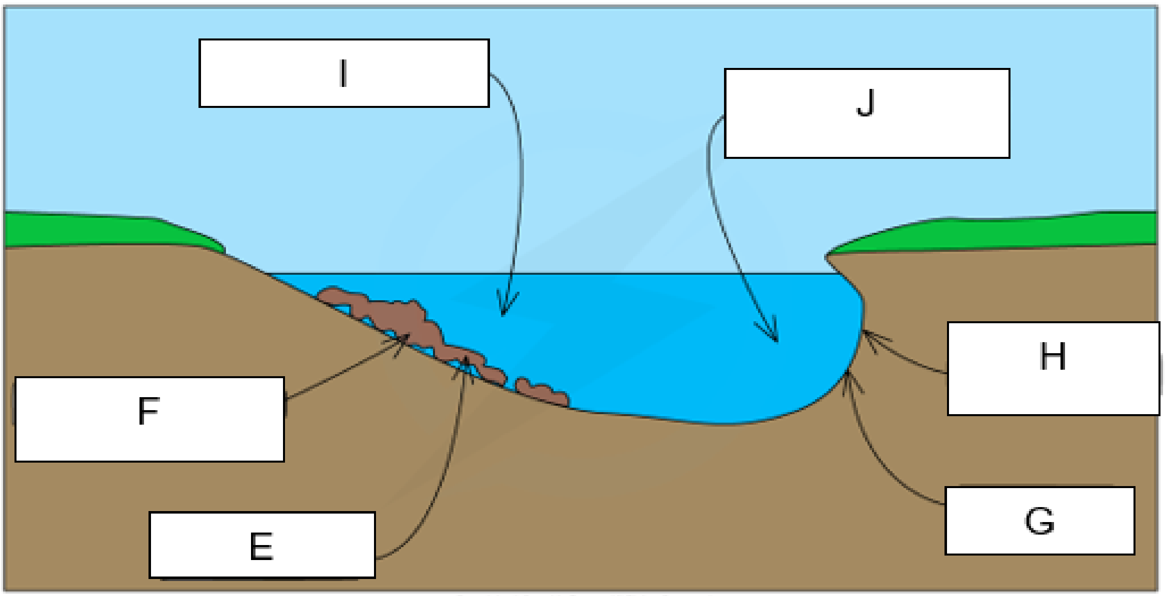

https://www.researchgate.net/figure/Map-of-the-Olifants-River-catchment-in-South-Africa

| COLUMN A | COLUMN B |
|---|---|
| 2.1.1 The stream order for the Olifants river as it flows into the Kruger National Park is a... | X. $2^{nd}$ order stream<br>Y. $3^{rd}$ order stream |
| 2.1.2 During times of drought the stream order will... | X. ...increase<br>Y. ...decrease |
| 2.1.3 The lower the stream order the … | X. ... smaller the drainage basin<br>Y. …bigger the drainage basin |
| 2.1.4 The higher the stream order the … | X. ... steeper the gradient of the river.<br>Y. … more gentle the gradient of the stream |
| 2.1.5 The Steelpoort and Spekboom tributaries are divided by … | X. ... the high lying area called a watershed.<br>Y. ... the high lying area called an interfluve. |

(5x1) (5)

**Question 2.2** **Meanders**

Various options are provided as possible answers to the following questions. Choose the answer and write only the letter (**A–D**) next to the question numbers (2.2.1 to 2.2.5) in the ANSWER BOOK, e.g. 2.2.6 D. Points **E - J** show different characteristics of the cross section of a meander.




2.2.1 The course of the river where one would find meanders forming.

A Upper course / Lower course

B Middle course / Lower course

C Middle course / Upper course

D All of the above

2.2.2 Name the process that is taking place at **E**.

A Headward erosion

B Deposition

C Erosion

D Infiltration

2.2.3 Name the process that is taking place at **G**.

A Headward erosion

B Deposition

C Erosion

D Infiltration

2.2.4 The river will flow the fastest at point ___ and slowest at point ___.

A I and J

B J and I

C None of the above. Fastest point is in the middle.

D None of the above. The speed of the river is uniform all over.

2.2.5 Once the meander neck erodes through, it forms a/an___ and when the feature dries out it becomes a/an ___

i meander pool.

ii ox bow lake.

iii dry ox bow lake.

iv meander scar.

A i and iv

B ii and iii

C ii and iv

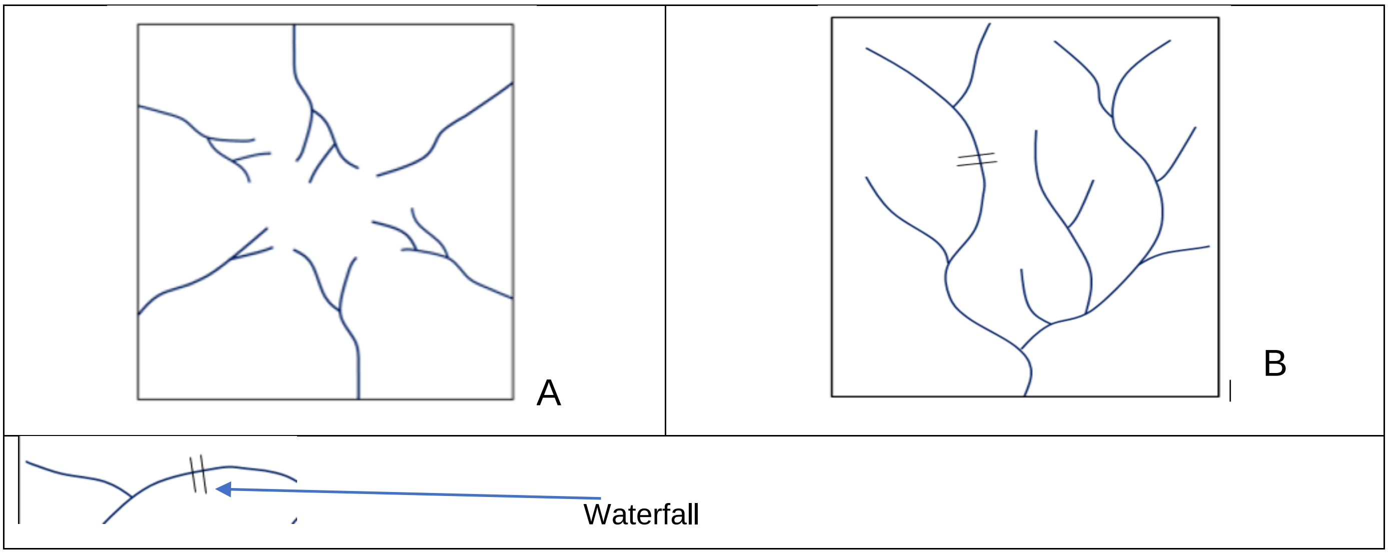
D i and iii

(5x1) (5)

| | |
|---|---|
| 2.1.1 Y (1) | 2.2.1 B.(1) |
| 2.1.2 Y (1) | 2.2.2 B (1) |
| 2.1.3 X (1) | 2.2.3 C (1) |
| 2.1.4 Y (1) | 2.2.4 B (1) |
| 2.1.5 X (1) | 2.2.5 C (1) |

**Question 2.3** **Drainage basins and characteristics.**

Refer to the diagrams below showing drainage patterns and answer the questions that follow.




2.3.1 Identify drainage patterns **A** and **B** respectively. (2x1) (2)

**A** Radial (1)

**B** Dendritic (1)

2.3.2 Differentiate between the underlying rock structure of drainage patterns **A** and **B** respectively. (2x2) (4)

**A** high lying hill or conical hill where the water/ river runs away from the highest point (2)

massive igneous rock structure, water running away from highest point (2)

underlying rock structure is massive/uniform/ equally resistant to erosion (2)

**B** uniform resistant and uniform gradient rock type/hardness (homogeneous) (2).

Underlying rock structure massive/ uniform (2)

Associated with massive igneous horizontal rock / sedimentary rocks (2)

2.3.3 Look at pattern **B**. the river shows an *ungraded* profile, and answer the following:

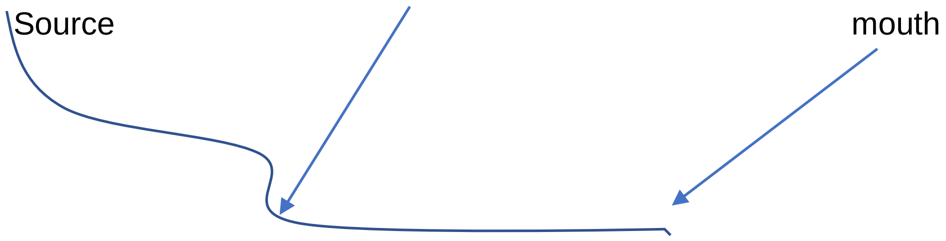
a) Give ONE reason to support the statement above. (1x1) (1)

There is a Knick point (1) / Waterfall on the profile (1)

b) Draw a longitudinal profile of river **B**. (1x1) (1)

drawing must show a waterfall (1)

ungraded profile (1)




c) Discuss the processes that would need to take place for the profile to become a graded profile. (2x2) (4)

More erosion would need to take place, eroding the Knick point (2)

River would need to become over graded to erode the waterfall away (2)

Headward erosion will take place causing the Knick point to move towards the source making the profile smooth (2)

hydraulic action will need to increase eroding the Knick point (2)

rejuvenation would need to take place to river can erode more(2)

2.3.4 Drainage pattern **B** shows a high drainage density.

a) Define the term *drainage density.* (1x2) (2)

The total length of streams (Km) over the area (in km $^2$) which it flows.(2)

total length of all the streams in km over the total area of the drainage basin km$^2$(2)

relationship between the total length of streams (km) and size of drainage basin (km$^2$) (2)

$\frac{total\ lenth\ of\ the\ stream\ (Km)}{Total\ drainage\ area\ (km2)}$ (2)

b) What is the relationship between drainage density and gradient? (1x1) (1)

The steeper the gradient the higher the density – more surface run off (1)

The gentler the gradient the lower the density- less surface run off (1)

**[15]**

**Question 2.4** **River Capture**

Refer to the figure, showing the Kouga River being captured by the Gamtoos river in the Eastern Cape.


2.4.1 Define the term *river capture*. (1x2) (2)

When one river captures the head waters from another river (2)

When one river cuts into the channel of another river and steals the headwaters (2)

When a more energetic river captures the headwaters of a less energetic river (2)

2.4.2 The process that caused the Kouga River to be captured by a tributary of the Gamtoos river, resulting in the Diep River, was abstraction.

Explain the process of abstraction. (1x2) (2)

The lowering and shifting of a watershed. (2)

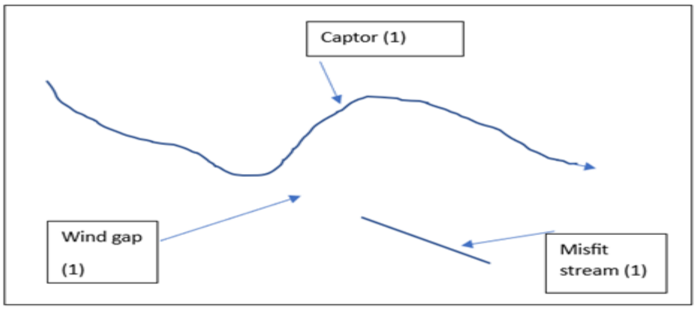
The steeper side of a watershed erodes quicker than the gentler side causing the watershed to shift backwards and become lower (2).

2.4.3 Draw a sketch of the river capture shown in the figure between the Kouga River and Diep River. On your sketch show the following:

Wind gap

Misfit stream

Captor stream (3x1) (3)



2.4.4 The Captor stream becomes rejuvenated.

a) Define the term river *rejuvenation.* (1x2) (2)

The increase in erosive power of a river due to increase volume and velocity (2)

is when a river has renewed energy and increased vertical erosion(2)

Is the process where rivers are re-energised to erode downwards (2) CONCEPT

b) List TWO landforms that can occur due to rejuvenation. (2x1) (2)

Knick point (1)

Incised meander (1)

Valley in a valley (1)

Waterfall(1)

River terrace (1)

c) Discuss what effect river capture will have on the volume of water along the Diep river and how this difference in volume will affect the commercial farmers along the river. (2x2) (4)

Volume will decrease as there will be less water (2)

Negative affect of the farmer and his production (2)

Less water farmers will need to pump water to the farm (2)

Increase in cost of product as farmers will need to build channels to get water to the farm (2)

**[15]**

**QUESTION 2 TOTAL: 40 MARKS**

**QUESTION 3** **Settlement**

**Question 3.1.**

Various possible answers are provided for each question. Write down **only the letter** of the correct answer next to the corresponding number.

3.1.1. In rural areas industrialisation and the modernisation of agricultural practices has led to:

A. Voluntary resettlement

B. Rural-urban migration

C. Basic need philosophy

D. Formal resettlement (1)

3.1.2. Which of the following factors is NOT a site factor to consider in the development of a settlement

A. Availability of water

B. Flat gradients

C. Central point of trade

D. A natural harbour (1)

3.1.3. Which is the smallest settlement?

A. Hamlet

B. Village

C. Isolated farmstead

D. City (1)

3.1.4. Which of the following would be an obstacle faced by rural-urban migrants?

A. Lack of capital

B. New lifestyles and associated costs

C. ONLY A

D. A and B (1)

3.1.5. Rural economic development is hindered because of:

A. These areas are often remote

B. These areas are seen as poor investments

C. Many workers suffer from diseases

D. All of the above (1)

**Question 3.2.**

State whether the following statements are **TRUE** or **FALSE**. Write only TRUE or FALSE next to the corresponding number in your answer book.

3.2.1. Rural areas are multifunctional in nature.

3.2.2. The situation of a settlement refers to its location in relation to its surrounding features.

3.2.3. Rural farming settlements is most prevalent in more economically developed countries.

3.2.4. Large scale solar farms would best be built in an urban area.

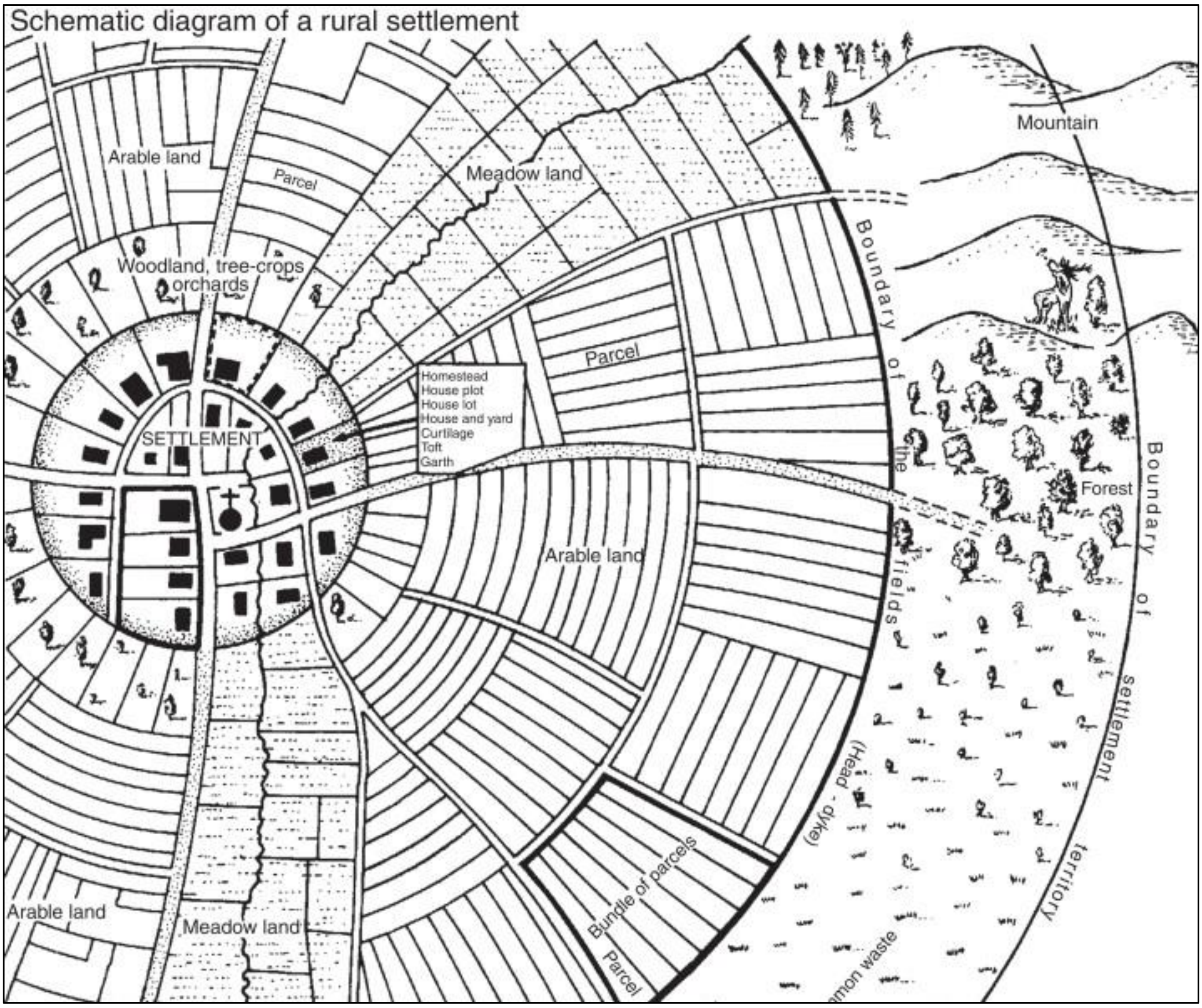
3.2.5. One of the causes of rural-urban migration is an increase in yield per unit area of agricultural land.

(5x1)(5)

| | |
|---|---|
| B | False |
| C | True |
| C | False |
| D | False |
| C | False |

**Question 3.3** **Rural Settlements**

Refer to the diagram below and answer the questions that follow.




3.3.1. Is this rural settlement likely to be engaging in commercial or subsistence farming? (1x1) (1)

Commercial (1)

3.3.2. Justify your answer to QUESTION 3.3.1. with TWO pieces of evidence from the diagram. (2x2) (4)

Large plots of land (2)

Organised road structure (2)

Clear farm boundaries (2)

No settlements within the farms (2)

3.3.3. This settlement developed on a wet-point site. Define the term *wet-point site.* (1x2) (2)

CONCEPT

A settlement that develops near a water source (2)

3.3.4. List THREE site factors that would have likely contributed to the development of this settlement. (3x1) (3)

Flat gradient

Good soil

Access to water

3.3.5. Define the concept of a settlement's *situation.* (1x2) (2)

The situation of a settlement refers to its location in relation to its surrounding features (2)

3.3.6. Is this settlement's situation likely to be positive or negative? Justify your answer. (1+2) (3)

Positive. (1)

This settlement appears very central, making it a likely trading port (2)

Having a proper river crossing present in the settlement makes it a central point for trade (2)

**[15]**

**Question 3.4** **Rural Settlement Issues**

Refer to the article below and answer the questions that follow.

# Rural healthcare in South Africa: A tale of adversity and heroism

*Although the state of rural healthcare in South Africa is fraught with challenges, including a shortage of staff, political interference, and a lack of resources, it is a story of adversity and heroism. Despite these challenges, there are a few islands of rural excellence where healthcare professionals have shown innovation and resilience, including the invention of apps to streamline referrals and the transformation of rural hospitals.*

## Heroism in adversity – SA's heart-rending rural health story

Public sector rural healthcare in South Africa is politically mismanaged, understaffed, underfunded and under-supervised, with unions reigning supreme, intimidating competent and suitably qualified leaders into shunning the top hospital jobs.

In an exclusive interview with Medbrief Africa, National Health Ombudsman, Professor Makgoba put abysmal public health care delivery in all but the Western Cape, Limpopo and KwaZulu Natal ("where they have at least found a little direction") down to dismal provincial and hospital leadership, infrastructural decay, and an almost universal lack of human resource capacity.

“Because of understaffing at every level, you have little institutional knowledge about the disciplines being practised in those hospitals. Everything has suffered, and everyone is overworked and overstretched. Seniors become irritated with the human resource leadership and infrastructure of the hospital,” he adds.

**Under 10% of graduates go rural.**

There are very few islands of rural excellence left in a sea of otherwise ideologically correct, semi-functional provincial health administrations. Despite a change in funding policy to get more money into rural areas (38 of the 52 health districts are considered rural), the money is almost always poorly or wastefully managed, while just 200 of the estimated annual 2 500 medical graduates are employed in State hospitals, (post community service), often not in rural areas – a function of ever-diminishing budgets.

**Administrative ‘guile’.**

In KwaZulu Natal, a new healthcare staff establishment norm was set in 2013 – but has never been implemented or translated into the Persal (human resources and payroll) system. This means the true vacancy rates for healthcare professionals, cleaners, kitchen staff, porters and admin staff are impossible to establish. The current standing instructions from KZN Department of Health are that if you cannot fill a post, you abolish it – a slashing of staff numbers via administrative guile.

“So, the vacancy rate will show as, say, twenty percent when it’s actually sixty percent,” Hobe adds.

Luckily, South Africa has its medical heroes and tireless advocates and activists who see the half-full glass. As Ben Gaunt, whose book, “Hope, A Goat and a Hospital”, chronicles his core team’s remarkable Zithulele journey, signs off on his e-mails, “If you don’t have a dream, how will you ever have a dream come true?”

*Guile - sly or cunning intelligence.

3.4.1. What rural issue is being highlighted in the article? (1x1) (1)

Lack of access to healthcare (1)

3.4.2. List TWO reasons for this issue as mentioned in the text. (2x1) (2)

“*shortage of staff, political interference, and a lack of resources” (2x1)*

3.4.3. Explain why it is challenging to get doctors working in rural areas. (2x2) (4)

Lack of funding (2)

Lack of Doctors (2)

Lack of recognition for Doctors (2)

Very remote areas (2)

Political interference means jobs are not available for them (2)

3.4.4. In a short paragraph of approximately 8 lines, discuss how you would solve the issue highlighted in the article. (4x2) (8)

Accept reasonable answers

Pay doctors who work in rural areas more

Provide benefits for working in these areas

Improve safety

Prioritise resources for rural areas

Improve infrastructure and accessibility in these areas

Provide tax benefits for private healthcare to develop there

Clamp down on corruption

4x2

**[15]**

**QUESTION 3 TOTAL: 40 MARKS**

# TOTAL SECTION A: 120 Marks

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# SECTION B: MAPWORK – Verulam

### QUESTION 4: MAP SKILLS AND CALCULATIONS

The questions below are based on the 1:50 000 topographic map (2931CA VERULAM), as well as the orthophoto map as part of the mapped area. Various options are provided as possible answers to the following questions. Write down the question number and the letter (A-D) only.

4.1.1 The co-ordinates of the Zwolle Estate at **H** ( A 2/ B2) are approximately…

A $31^0$ 36’ 0” S $29^0$ 03’ 37” E

B $29^0$ 36’ 0” S $31^0$ 03’ 37” E

C $31^0$ 36’ 37” S $29^0$ 03’ 0” E

D $29^0$ 36’ 37” S $31^0$ 03’ 0” E

4.1.2 The topographic map code directly West of 29 31 CA is…

A 29 31 DD

B 29 30 DB

C 29 31 DB

D 29 30 DD

4.1.3 On the orthophotograph, the gradient between spot height 29 (**A5**) and spot height 133 (**D2**) is ………..and is classified as a ……….. slope.

i 1: 76.9

ii 1: 15.8

iii steep

iv gentle

A i and iii

B ii and iii

C i and iv

D ii and iv

4.1.4 The scale of the topographic map is 5 times … than the orthophoto, and the topographic map covers a … area.

A larger; smaller

B larger; larger

C smaller; larger

D smaller; smaller (4x1) (4)

| |
|---|
| **B** |
| **D** |

| |
|---|
| **B** |
| **C** |

4.1.5 On the topographical map, work out the magnetic bearing of the benchmark 52.8 (**E5**)

***from*** trig beacon 70 (**C3**).

Difference in years: 7 years

Total annual change: 9' x 7 = 63' ($1^0$ 03'W)

**Magnetic Bearing = True Bearing + Magnetic Declination** (4x1) (4)

True bearing = $127^0$ (1) ($126^0 – 128^0$)

Declination = $25^0$ 04' + $1^0$ 03'W = $26^0$ 07' West of true north (2)

Magnetic bearing = $153^0$ 07' (1) ($152^0$ 07' – $154^0$ 07')

4.1.6 Calculate the area of the demarcated red orthophoto, on the topographical map, in **$km^2$.**
Use the following length distance: 3.7 cm = 1.85km

**Formula: Area = Length x Breadth** (2x1) (2)

Breadth : 3.2cm ( 3.1cm – 3.3cm) = 1.6km (1) ( 1.55km – 1.65km)

Area = Length x Breadth (2x1) (2)

1.85km x 1.6km

= 2.96$km^2$ (1) (2.87$km^2$- 3.05$km^2$)

**[10]**

4.2 **APPLICATION & INTERPRETATION**

The questions below are based on the 1:50 000 topographic map (2931CA VERULAM), as well as the orthophoto map as part of the mapped area. Various options are provided as possible answers to the following questions. Write down the question number and the letter (A-D) only.

4.2.1 The river feature found at **J** is the result of …., on the topographical map.

A erosion

B deposition

C undercutting

D abstraction

4.2.2 The King Shakas landing strips (**C5**) are running………………..due to the strong ……..winds

i North west to southeast

ii South west to northeast

iii South easterly

iv North westerly

A i and iii

B ii and iii

C i and iv

D ii and iv (2x1) (2)

| |
|---|
| B<br>B |

4.2.3 Refer to Bellamont (**E5**) on the topographical map.

a) Is the settlement pattern at Bellamont *dispersed* or *nucleated*? (1x1) (1)

*dispersed*(1)

b) State TWO site factors that favoured the development of this settlement. (2x1) (2)

Gentle gradient (1)

Fertile soil (1)

Water access (1)

Accessibility (1)

c) Provide the situation of Bellamont? (1x2) (2)

Good trading opportunities (2)

Just off the main road (2)

South of the Moloti river (2)

Near the town of Verulum ( Eastern side ) (2)

South of the King Shaka international airport (2)

4.2.4 Bellamont settlement is linked to commercial maize farming.

a) Would this type of farming be *commercial* or *subsistence*? (1x1) (1)

Commercial (1)

b) Support your answer in QUESTION 3.2.4 a, using evidence from the map. (1x2) (2)

Very large piece of farmed land (2)

Not many other farms close by (2)

4.2.5 Zwolle Estate at **H** (**A/B2**) would be classified as *dry point* settlement.

Give a reason to support this statement. (1x2) (2)

Situated up on the hill away from the water or the Moloti river (2)

Built above the flood plain (2)

Far away from a water source (2)

This settlement is trying to avoid risk of flooding (2)

Farm built on higher ground to avoid flooding (2)

Farm built away from the river to avoid flooding (2)

**[12]**

### 4.3 GEOGRAPHICAL INFORMATION SYSTEMS

Refer to the information below and answer the questions that follow.

ARC • LNR

*Excellence in Research and Development*

**About this app**

As quick warning measures and improved technologies are more at demand for improved maize production and food security, the ARC (Agricultural Research Council) is pleased to introduce the maize producers to new and improved mobile app that comprises of the latest collection of scientifically proven data on maize production, which can be easily downloaded for quick and updated information on the go.

The ARC (Agricultural Research Council) remains committed in providing the best technologies to the maize farmers

GIS is made up of many different components, such as humans, hardware and software.

4.3.1 What type of component will ARC.LNR be part of? (1x1) (1)

Software (1)

4.3.2 List TWO different types of data that ARC.LNR could offer the farmers? (2x1) (2)

weather reports (1)

Flood warnings (1)

Frost/ snow warnings (1)

What time of the year is best suited to plant maize (1)

Genetically modified seeds (1)

Drought warnings (1)

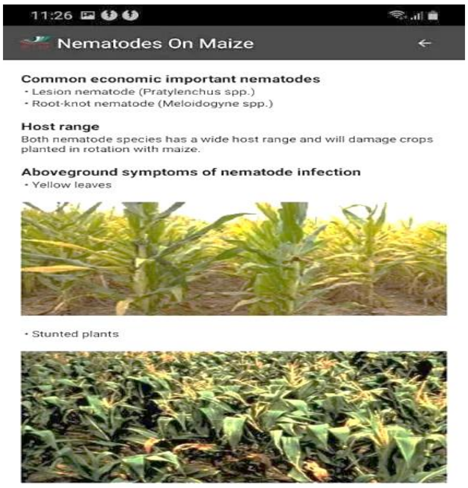
How to save water (1)

Best place to sell product (1)

What price maize is selling at (1)

Use your discretion ….

4.4 The following information was taken from the ARC.LNR farming site.




4.4.1 Classify the above photograph as either primary or secondary data. (1x1) (1)

Primary (1)

4.4.2 Resolution is an important factor when taking photographs.

a) Define the term *resolution*. (1x2) (2)

The detail with which a photo graph depicts the location, shape or geographical features (2)

The total number of pixels a photograph has to make it clear (2)

The ability of a remote sensing sensor to create a sharp and clear image. (2)

b) How would a higher resolution in the photographs assist a GIS specialist to find a solution for the environmental issue depicted. (1x2) (2)

View what the problem is as shown in the picture (2)

The clearer the picture the more help and assistance can be given (2)

The clearer the picture the easy it will be to distribute for discussion (2)

**[8]**

**QUESTION 4 TOTAL: 30 MARKS**

## TOTAL SECTION B: 30 Marks

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# <u>GRAND TOTAL: 150 Marks</u>